

OPC DA 自主采集操作手册

LONGNAME → TagID → ReadList 全链路

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2026 年 7 月 13 日

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1 概述

本手册覆盖油田作业区 IO 服务器 292 台设备、46,046 个测点的完整 OPC DA 数据采集方案。

1.1 适用范围

- 目标: pSpace 6.0 实时数据库 (11.66.12.130:8889)
- 环境: 32 位 Python 3.11 + psAPISDK.dll
- 数据源: Oracle PROJECT_TAGPAR → parse.db tag_definition

1.2 采集链路

Step 1: parse.db 查 LONGNAME

```
SELECT longname FROM tag_definition  
WHERE driver_name='OPC_FC_Client' AND device_id='DX1ZRZ'  
→ /CY1C8K/DX1ZRZ/ZD010833ALA172
```

Step 2: psAPI 解析 TagID

```
psAPI_Tag_GetIdByLongName(handle, LONGNAME, &tagId)  
→ TagID = 11945
```

Step 3: 读取实时值

```
psAPI_Real_ReadList(handle, 1, [11945], &values, &errors)  
→ value = 2.2017, timestamp = 2026-07-13 22:01:10, quality = GOOD
```

2 环境准备

2.1 32 位 Python

```
C:\Python311-32\python.exe --version  
# Python 3.11.9 (32-bit)
```

2.2 依赖文件

- psAPISDK.dll + 所有依赖 DLL (IO ServerOnLine 目录)
- parse.db (本地 tag_definition 表, 46,046 行)

2.3 网络要求

- 可达 11.66.12.130:8889 (pSpace)
- Oracle 可通过 WinRM 131 中转 (可选, 仅需导入时)

3 设备清单查询

3.1 按驱动查询设备

```
import sqlite3
db = sqlite3.connect('data/parse.db')

# 列出所有驱动
for row in db.execute("""
SELECT driver_name, COUNT(DISTINCT device_id) as devs, COUNT(*) as
tags
FROM tag_definition GROUP BY driver_name
"""):
    print(f"{row[0]}: {row[1]} devices, {row[2]} tags")

# 列出某驱动的设备
for row in db.execute("""
SELECT DISTINCT device_id, well_name
FROM tag_definition WHERE driver_name='OPC_FC_Client'
LIMIT 10
"""):
    print(f"{row[0]}->{row[1]}")
```

3.2 按设备查询 LONGNAME

```
device = 'DX1ZRZ'
for row in db.execute("""
SELECT tag_name, longname, data_type, range_max, range_min, unit
FROM tag_definition
WHERE device_id=?
LIMIT 10
""", (device,)):
    print(f"{row[0]:30s} {row[1]:50s} [{row[4]}-{row[3]}] {row[5]}")
```

4 LONGNAME → TagID 解析

4.1 标准工具

```
C:\Python311-32\python.exe tools/pspace_collector.py --ids "11945,5657"
```

4.2 批量解析

```
import ctypes, os
os.add_dll_directory(r'D:\ai\dgiot_lite\io服务器分析\IO_ServerOnline')
dll = ctypes.CDLL('psAPISDK.dll')

def resolve_tags(paths):
    dll.psAPI_Common_StartAPI.restype = ctypes.c_int32
    dll.psAPI_Common_StartAPI()
    dll.psAPI_Server_Connect.argtypes = [ctypes.c_char_p]*3 + [ctypes.
        POINTER(ctypes.c_uint16)]
    dll.psAPI_Server_Connect.restype = ctypes.c_int32
    h = ctypes.c_uint16(0xFFFF)
    dll.psAPI_Server_Connect(b'11.66.12.130:8889', b'admin', b'
        DQYTA11_pass', ctypes.byref(h))

    dll.psAPI_Tag_GetIdByLongName.argtypes = [ctypes.c_uint16, ctypes.
        c_char_p, ctypes.POINTER(ctypes.c_uint32)]
    dll.psAPI_Tag_GetIdByLongName.restype = ctypes.c_int32

    results = {}
    for path in paths:
        tid = ctypes.c_uint32(0)
        r = dll.psAPI_Tag_GetIdByLongName(h, path, ctypes.byref(tid))
        if r == 0:
            results[path.decode()] = tid.value

    dll.psAPI_Server_Disconnect(h)
    dll.psAPI_Common_StopAPI()
    return results
```

5 实时数据读取

5.1 单次读取

```
def read_tags(tag_ids):
    dll = ctypes.CDLL('psAPISDK.dll')
    dll.psAPI_Common_StartAPI.restype = ctypes.c_int32
    dll.psAPI_Common_StartAPI()
    dll.psAPI_Server_Connect.argtypes = [ctypes.c_char_p]*3 + [ctypes.
        POINTER(ctypes.c_uint16)]
    dll.psAPI_Server_Connect.restype = ctypes.c_int32
    h = ctypes.c_uint16(0xFFFF)
    dll.psAPI_Server_Connect(b'11.66.12.130:8889', b'admin', b'
        DQYTA11_pass', ctypes.byref(h))

    dll.psAPI_Real_ReadList.argtypes = [ctypes.c_uint16, ctypes.
        c_uint32, ctypes.c_void_p, ctypes.c_void_p, ctypes.c_void_p]
    dll.psAPI_Real_ReadList.restype = ctypes.c_int32

    n = len(tag_ids)
    ids_arr = (ctypes.c_uint32 * n)(*tag_ids)
    ppV = ctypes.c_void_p(0)
    r = dll.psAPI_Real_ReadList(h, n, ids_arr, ctypes.byref(ppV), 0)

    results = []
    if r == 0 and ppV.value:
        raw = ctypes.string_at(ppV.value, n * 24)
        for i, tid in enumerate(tag_ids):
            import struct
            v = struct.unpack('<f', raw[i*24+16:i*24+20])[0]
            ts = struct.unpack('<I', raw[i*24:i*24+4])[0]
            quality = raw[i*24+20]
            results.append({'tag_id': tid, 'value': v, 'timestamp': ts,
                'quality': quality})

    dll.psAPI_Server_Disconnect(h)
    dll.psAPI_Common_StopAPI()
    return results
```

5.2 PS_DATA 内存布局

6 常用 SQL 查询

7 数据覆盖

8 限制与注意事项

- 单次连接限制: 每次 Connect 只能执行一次 ReadList (最多约 20 个 Tag)
- 解析和读取分开: Tag_GetIdByLongName 和 Real_ReadList 必须在不同连接中
- 限流: 连续连接间隔 > 5 秒, 避免 -19998
- 32 位必须: psAPISDK.dll 仅支持 32 位 Python
- 网络依赖: 需可达 11.66.12.130:8889
- 离线可用: parse.db 本地查询无需 Oracle

9 故障排查

偏移	大小	字段
0-3	4B	Time.Seconds (uint32 LE)
4-7	4B	Time.NanoSec
8-11	4B	DataType
12-15	4B	(reserved)
16-19	4B	Float Value
20-23	4B	Quality (0xC0=GOOD)

场景	SQL
查设备所有 Tag	SELECT * FROM tag_definition WHERE device_id='DX1ZRZ'
按驱动统计	SELECT driver_name, COUNT(*) FROM tag_definition GROUP BY 1
查某 Tag 的 LONGNAME	SELECT longname FROM tag_definition WHERE tag_name LIKE '%ALA172'
查量程范围	SELECT device_id, tag_name, range_min, range_max, unit FROM tag_definition WHERE driver_name='OPC_FC_Client' LIMIT 10

驱动	设备	Tag	访问	状态
OPC_FC_Client (Kepware)	34	40,956	LONGNAME→TagID→ReadList	已验证
IM_A11_RTU (油田)	161	3,638	同上	已验证
Standard_Umodbus	50	1,452	同上	已验证
保护装置	47	—	ReadList 直读	已验证

错误码	含义	解决
-19998	未初始化	重新 Connect 获取新 handle
-19997	Tag 数超限	减少至 20 个以内
-18591	密码错误	确认凭据
-18592	用户不存在	确认用户名
-19391	Tag 不存在	检查 LONGNAME 是否正确